Nathan Houck

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EXECUTIVE SUMMARY

Aerospace engineering senior (Astronautics) with hands-on experience designing, integrating, and testing mechanical and thermal systems; integration engineer on a CubeSat design team. Demonstrated ability to lead technical teams through full development cycles. Seeking an entry-level systems engineering position.

EDUCATION

Embry-Riddle Aeronautical University Expected May 2027
B.S. Aerospace Engineering, Astronautics Concentration | GPA: 3.72/4.0 *Daytona Beach, FL*
Honors: Dean's List (5 semesters); Honor Roll (Fall 2024); Exceptional Eagles Award, ERAU Student Involvement Awards (2025)
Relevant Coursework: Spacecraft Systems, Spacecraft Attitude Dynamics, Space Propulsion, Aerospace Structures, Experimental Aerodynamics Lab, Computing for Engineers

WORK EXPERIENCE

EPI-USE America May 2026 - Sep. 2026
AI Exploration Intern *Remote*
- Compared build-versus-buy options for software on cost, upkeep, and integration, and wrote recommendation briefs practice leaders used to make decisions
- Designed a Python tool to scrape public procurement portals through a unified interface, ranking results into one spreadsheet and finding **5x more opportunities**, each more relevant to the company's expertise
- Built an AWS workflow that splits a proposal request into individual requirements and scores how well the company meets each one, rejecting the bid if a required item is missed

Embry-Riddle Aeronautical University Jan. 2026 - Aug. 2026
Grading Assistant, AE 319 Aerodynamics *Daytona Beach, FL*

DESIGN PROJECTS

Project SENTINEL, Senior Spacecraft Design Aug. 2026 - Present
Integration Engineer *ERAU*
- Coordinate **five subteams** on a 3U CubeSat that autonomously catalogs RSOs in the GEO belt
- Write the CONOPS document from the mission needs document to set design requirements
- Derive requirements from the CONOPS and mission needs documents ahead of an SRR presentation

Autonomous Maritime Robotics Association (RoboSub) Aug. 2024 - Aug. 2025
Heat Transfer Team Lead; Mechanical Engineering Team *ERAU*
- Led a study of cooling options (heat pipes, thermoelectric coolers, PCMs) to cut the ~190°F hull peak by **over 60°F**
- Designed and built an improved thruster mount with three teammates, and a torpedo launcher and dropper

LABS & RESEARCH

Aerospace Structures and Instrumentation Lab (AE 416/417) Aug. 2026 - Present
Structural Testing and Instrumentation *ERAU*
- Install strain gages on test specimens and check static load test results against theory

Experimental Aerodynamics Laboratory (AE 314/315) Jan. 2026 - Apr. 2026
Wind Tunnel and Supersonic Tunnel Testing *ERAU*
- Ran six wind tunnel and supersonic tunnel experiments start to finish: setup, data collection, MATLAB data reduction, uncertainty analysis
- Located a shock in an overexpanded supersonic nozzle by comparing **16 pressure readings** with Schlieren images, and designed a boundary-layer tripping experiment on a NACA 0012 airfoil

Thermal Science Laboratory, with Instituto Tecnológico de Aeronáutica (Brazil) Aug. 2025 - Dec. 2025
Undergraduate Researcher, Phase Change Material Testing *ERAU*
- Designed a tilting test section in Inventor, with one heated wall and one fixed temperature wall, to study the behavior of PCMs

SKILLS

Systems Engineering: Requirements, interfaces, trade studies, verification, technical writing
Test & Measurement: Test planning, wind tunnel testing, strain gages, uncertainty analysis
Programming: Python, MATLAB, SQL, Java, C/C++ (basic), Git, AWS
Engineering Software: MATLAB, CATIA V5, Autodesk Fusion and Inventor, 3D printing, technical drafting

LEADERSHIP/INVOLVEMENT

Conservation Officer, Studying Habitats and Reef Kingdoms (2026 - Present): reef and mangrove programs
Member, Experimental Jet Engine Propulsion Club | **Flight Student** (28.1 hrs) | Science Olympiad